WKB 数值格式与 AP 理论分析技术路线图

1 当前工作的定位

当前工作的核心目标不是完整证明全离散 AP 定理，而是建立一个面向稀有突变极限的 WKB 数值格式框架。主要任务包括格式构造、正性保持、重构密度方程推导、条件型稳定判据以及数值验证。

这篇文章应当收缩为如下主线

WKB 格式构造 $\Longrightarrow$ 正性保持 $\Longrightarrow$ 重构密度方程 $\Longrightarrow$ 条件型密度稳定性 $\Longrightarrow$ 数值验证.

其中完整的 uniform-in-ε AP 稳定性证明不在本文中展开，而作为后续理论工作的主要方向。

2 当前文章主线

当前文章建议采用如下技术路线。

1. 建立 WKB 重构
$$n^{\varepsilon}_{j,k} = W^{\varepsilon}_{j,k} E^{\varepsilon}_{k}, \qquad E^{\varepsilon}_{k} = \exp\left(\frac{u^{\varepsilon}_{k}}{\varepsilon}\right).$$
2. 区分分析用质量与实际重构密度
$$m^{\varepsilon}_{j} = \Delta\theta \sum_{k=1}^{K} n^{\varepsilon}_{j,k}, \qquad \rho^{\varepsilon}_{j,Q} = Q^{\varepsilon}_{\theta}[W^{\varepsilon}_{j,\cdot}, u^{\varepsilon}].$$
3. 推导重构密度方程
$$\varepsilon \frac{d}{dt} n^{\varepsilon}_{j,k} - D_k \delta_x^2 n^{\varepsilon}_{j,k} = (K_j - \rho^{\varepsilon}_{j,Q}) n^{\varepsilon}_{j,k} + R^{\varepsilon}_{j,k}.$$
4. 证明正性保持
$$W^{\varepsilon}_{j,k} \geq 0, \qquad n^{\varepsilon}_{j,k} \geq 0.$$
5. 引入条件型稳定判据。若
$$\rho^{\varepsilon}_{j,Q} \geq c_Q m^{\varepsilon}_{j} - e_Q$$
且
$$\mathcal{R}^{\varepsilon,D}_{j} \leq C_R m^{\varepsilon}_{j} + r_h,$$
则可以恢复类似连续情形的密度上界。

6. 说明该结论是 conditional stability criterion，而不是完整 AP theorem。
7. 通过数值实验验证 WKB 方法在小 ε 下能够稳定捕捉 fittest trait，尤其展示 Laplace-type reconstruction 相比 direct quadrature 的优势。

3 后续完整 AP 理论路线

完整 AP 证明可作为后续独立工作，核心是闭合如下 bootstrap 链条

$$\text{初值准备} \Longrightarrow m^\varepsilon, \rho_Q^\varepsilon \text{ 有界} \Longrightarrow u^\varepsilon \text{ 离散 Lipschitz 估计} \Longrightarrow W^\varepsilon \text{ 加权差分估计} \Longrightarrow \mathcal{R}^{\varepsilon,D} \text{ 一侧控制} \Longrightarrow \text{完整 AP}$$

其中最困难的部分是离散相变量的 Bernstein 型估计和 trait-direction remainder 的闭合控制。

4 技术路线图

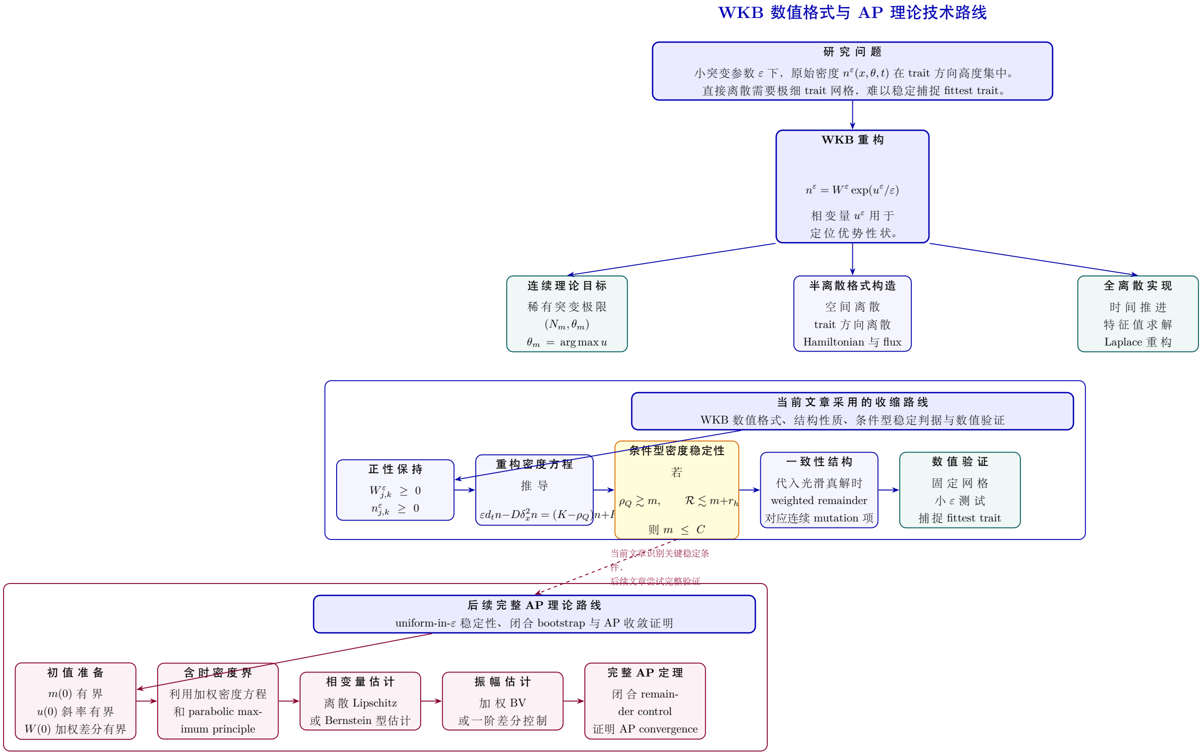

图 1: 当前 WKB 数值方法文章与后续完整 AP 理论分析的技术路线。

5 后续理论工作的关键难点

后续完整 AP 理论需要解决以下关键问题。

5.1 一侧 remainder control 的验证

当前文章中使用的条件型稳定判据为

$$\mathcal{R}_j^{\varepsilon,D} \le C_R m_j^\varepsilon + r_h.$$

后续需要从格式结构出发证明该估计，而不是将其作为假设。困难在于 WKB 离散后不再具有精确链式法则，$R_{j,k}^\varepsilon$ 中同时包含

$$\varepsilon^2 E_k^\varepsilon \delta_\theta^2 W_{j,k}^\varepsilon, \qquad \varepsilon n_{j,k}^\varepsilon \delta_\theta^2 u_k^\varepsilon, \qquad n_{j,k}^\varepsilon \widehat{H}, \qquad \varepsilon E_k^\varepsilon \mathcal{D}_\theta \widehat{\mathcal{F}}.$$

这些项不能简单逐项控制，需要结合相变量和振幅变量的稳定性。

5.2 离散相变量估计

连续理论中，u^ε 的 Lipschitz 估计通常通过 Bernstein 方法获得。离散情形更困难，原因包括

缺少精确链式法则, 数值 Hamiltonian 可能非光滑, 常数可能依赖网格尺度.

后续可以考虑两条路线。

第一条是直接对离散 HJ 方程建立差分梯度估计。第二条是通过某种特征值型相变量构造，先得到正特征向量，再取对数得到相函数。

5.3 含时 bootstrap 闭合

更自然的完整证明路线是含时 bootstrap。设初值 well-prepared，并假设在 $[0,T_*]$ 上已有粗略界

$$m^\varepsilon \le M, \qquad |D_\theta^\pm u^\varepsilon| \le L, \qquad \text{weighted variation of } W^\varepsilon \le M_W.$$

然后依次证明更好的密度界、相变量估计、振幅估计和 remainder control。若这些估计可以闭合，则可推出 $T_* = T$，从而得到 uniform-in-ε 稳定性。

6 当前文章建议保留的表述

当前文章中建议使用如下措辞。

> The present analysis should be understood as an AP-oriented structural analysis rather than a complete uniform-in-ε convergence theorem. We identify the key

one-sided remainder condition under which the discrete density estimate can be recovered. A full AP theorem would require additional stability estimates for the phase and amplitude variables and is left for future work.

7 部分移除的性质

Lemma 7.1 (Local-in-time discrete bounds for the phase)**.** *Assume that*

$$\max_k u_k^\varepsilon(0) \le C_0\varepsilon, \qquad \max_k \Big(|D_\theta^- u_k^\varepsilon(0)| + |D_\theta^+ u_k^\varepsilon(0)|\Big) \le L_0,$$

and that the initial density $\rho_j^\varepsilon(0)$ is bounded in space. Then there exist $T > 0$, possibly depending on ε, and constants C_T, $L_T > 0$ such that

$$\max_{t\in[0,T]} \max_k u_k^\varepsilon(t) \le C_T\varepsilon, \qquad \max_{t\in[0,T]} \max_k \Big(|D_\theta^- u_k^\varepsilon(t)| + |D_\theta^+ u_k^\varepsilon(t)|\Big) \le L_T.$$

证明*.* The semi-discrete equations form a finite-dimensional ODE system with locally Lipschitz right-hand side. Hence local solutions exist for each fixed ε. Continuity in time and boundedness of the discrete source term $\mathcal{H}_k(\boldsymbol{\rho}(t))$ on a short time interval imply local bounds for the one-sided trait slopes of u^ε. The phase equation (**??**) then gives a local bound for $\partial_t u_k^\varepsilon$, which yields the stated estimate after possibly shrinking the time interval. □

Lemma 7.2 (Local-in-time weighted first-difference bound for the amplitude)**.** *Assume that $W_{j,k}^\varepsilon(0) \ge 0$ for all j and k. Then there exists $T > 0$ such that, for all $t \in [0,T]$ and all j,*

$$\Delta\theta \sum_{k=1}^{K} \exp\big(u_k^\varepsilon(t)/\varepsilon\big)\, |\delta_\theta^+ W_{j,k}^\varepsilon(t)| \le M_T\, \rho_{\varepsilon,j}(t), \qquad \delta_\theta^+ W_{j,k}^\varepsilon = \frac{W_{j,k+1}^\varepsilon - W_{j,k}^\varepsilon}{\Delta\theta}. \tag{1}$$

证明*.* Set $g_k(t) = \exp(u_k^\varepsilon(t)/\varepsilon)$. By Lemma **??**, we have $W_{j,k}^\varepsilon(t) \ge 0$ on the local existence interval. Hence

$$|W_{j,k+1}^\varepsilon(t) - W_{j,k}^\varepsilon(t)| \le W_{j,k+1}^\varepsilon(t) + W_{j,k}^\varepsilon(t).$$

Summing against the weights $g_k(t)$ produces two terms. One is exactly $\rho_{\varepsilon,j}(t)$. The other is controlled by comparing neighboring weights, which is possible thanks to the local slope bound for u^ε from Lemma 7.1. This yields (1). □

Proposition 7.1 (Conditional continuation criterion)**.** *Fix $T > 0$. Assume that a semi-discrete solution $(u^\varepsilon, W^\varepsilon)$ of (**??**)–(**??**) satisfies*

$$0 \le \rho_j^\varepsilon(t) \le \overline{\rho}_T, \qquad \forall\, j = 1,\dots,J,\ t \in [0,T],$$

for some constant $\overline{\rho}_T$ independent of ε. Assume moreover that $D(\theta)$ is Lipschitz and that the discrete principal eigenvalue $\mathcal{H}_k(\boldsymbol{\rho})$ depends Lipschitz-continuously on θ_k whenever $\boldsymbol{\rho}$ stays in *bounded sets. Then the local bounds of Lemmas 7.1 and 7.2 extend to the whole interval* $[0,T]$*. In particular, there exist constants* C_T*,* L_T*, and* M_T*, independent of* ε*, such that*

$$\max_{t\in[0,T]}\max_k u_k^\varepsilon(t)\le C_T\varepsilon,\qquad \max_{t\in[0,T]}\max_k\Big(|D_\theta^- u_k^\varepsilon(t)|+|D_\theta^+ u_k^\varepsilon(t)|\Big)\le L_T,$$

and

$$\Delta\theta\sum_{k=1}^K \exp\big(u_k^\varepsilon(t)/\varepsilon\big)\,|\delta_\theta^+ W_{j,k}^\varepsilon(t)|\le M_T\,\rho_{\varepsilon,j}(t),\qquad \forall\, t\in[0,T],\ \forall\, j.$$

证明. The proof is a continuation argument. The local estimates in Lemmas 7.1 and 7.2 depend only on boundedness of the coefficients in the ODE system. Once ρ^ε is bounded on $[0,T]$, the discrete effective Hamiltonian remains bounded and Lipschitz with constants depending only on T and the data. The local estimates can therefore be restarted from any intermediate time and continued to the whole interval. □

Lemma 7.3 (Consistency of the weighted trait remainder evaluated at the exact solution)**.** *Let* $(u^\varepsilon, W^\varepsilon)$ *be the exact solution pair of the continuous WKB system* (**??**)*, and define*

$$n^\varepsilon(x,\theta,t)=W^\varepsilon(x,\theta,t)e^{u^\varepsilon(\theta,t)/\varepsilon}.$$

Let $\{\theta_k\}_{k=1}^K$ *be a uniform periodic grid with step size* $\Delta\theta$*, and define*

$$u_k^\varepsilon(t)=u^\varepsilon(\theta_k,t),\qquad W_{j,k}^\varepsilon(t)=W^\varepsilon(x_j,\theta_k,t),\qquad n_{j,k}^\varepsilon(t)=n^\varepsilon(x_j,\theta_k,t).$$

Let $D_k = D(\theta_k)$*,* $\alpha_k = 1/D_k$*, and let* $R_{j,k}^\varepsilon(t)$ *be defined by* (**??**) *with the same numerical Hamiltonian* $\widehat{H}$ *and numerical flux* $\widehat{F}$ *as in the scheme. Define the weighted aggregate remainder*

$$\mathcal{R}_j^{\varepsilon,D}(t):=\Delta\theta\sum_{k=1}^K\alpha_k R_{j,k}^\varepsilon(t).$$

Assume that $D\in W^{2,\infty}(\mathbb{T})$ *with* $D\ge D_{\min}>0$*, that*

$$u^\varepsilon(\cdot,t)\in W^{3,\infty}(\mathbb{T}),\qquad W^\varepsilon(x_j,\cdot,t)\in W^{2,\infty}(\mathbb{T}),\qquad t\in[0,T],$$

and that $\widehat{H}$ *and* $\widehat{F}$ *are consistent first-order monotone approximations of their continuous counterparts and are Lipschitz in their arguments. Then, for each fixed* $\varepsilon>0$*, there exists a constant* $C_{\text{cons}}(\varepsilon)>0$*, independent of* $\Delta\theta$*, such that*

$$\left|\mathcal{R}_j^{\varepsilon,D}(t)-\varepsilon^2\,\Delta\theta\sum_{k=1}^K n_{j,k}^\varepsilon(t)\,\delta_\theta^2\Big(\frac{1}{D_k}\Big)\right|\le C_{\text{cons}}(\varepsilon)\,\Delta\theta,\qquad t\in[0,T]. \tag{2}$$

In particular, the leading term vanishes when D *is constant.*

证明. Since the exact solution is smooth in the trait variable, the discrete difference operators are consistent when evaluated at the sampled exact solution. Using the consistency and Lipschitz continuity of $\widehat{H}$ and $\widehat{F}$, we obtain pointwise consistency relations for the numerical Hamiltonian and numerical flux. Comparing (**??**) with the continuous identity that the trait contribution equals $\varepsilon^2\partial_{\theta\theta}n^\varepsilon$, we obtain

$$R^\varepsilon_{j,k}(t) = \varepsilon^2\delta^2_\theta n^\varepsilon_{j,k}(t) + \tau^\varepsilon_{j,k}(t), \qquad |\tau^\varepsilon_{j,k}(t)| \le C_{\mathrm{cons}}(\varepsilon)\Delta\theta.$$

Multiplying by α_k and summing over k, we get

$$\mathcal{R}^{\varepsilon,D}_j(t) = \varepsilon^2\Delta\theta\sum_{k=1}^{K}\alpha_k\delta^2_\theta n^\varepsilon_{j,k}(t) + \Delta\theta\sum_{k=1}^{K}\alpha_k\tau^\varepsilon_{j,k}(t).$$

Periodic summation by parts yields

$$\Delta\theta\sum_{k=1}^{K}\alpha_k\delta^2_\theta n^\varepsilon_{j,k} = \Delta\theta\sum_{k=1}^{K}n^\varepsilon_{j,k}\delta^2_\theta\alpha_k,$$

which gives the leading term in (2). The error term is bounded by $C_{\mathrm{cons}}(\varepsilon)\Delta\theta$ because $\alpha_k \le D^{-1}_{\min}$. □